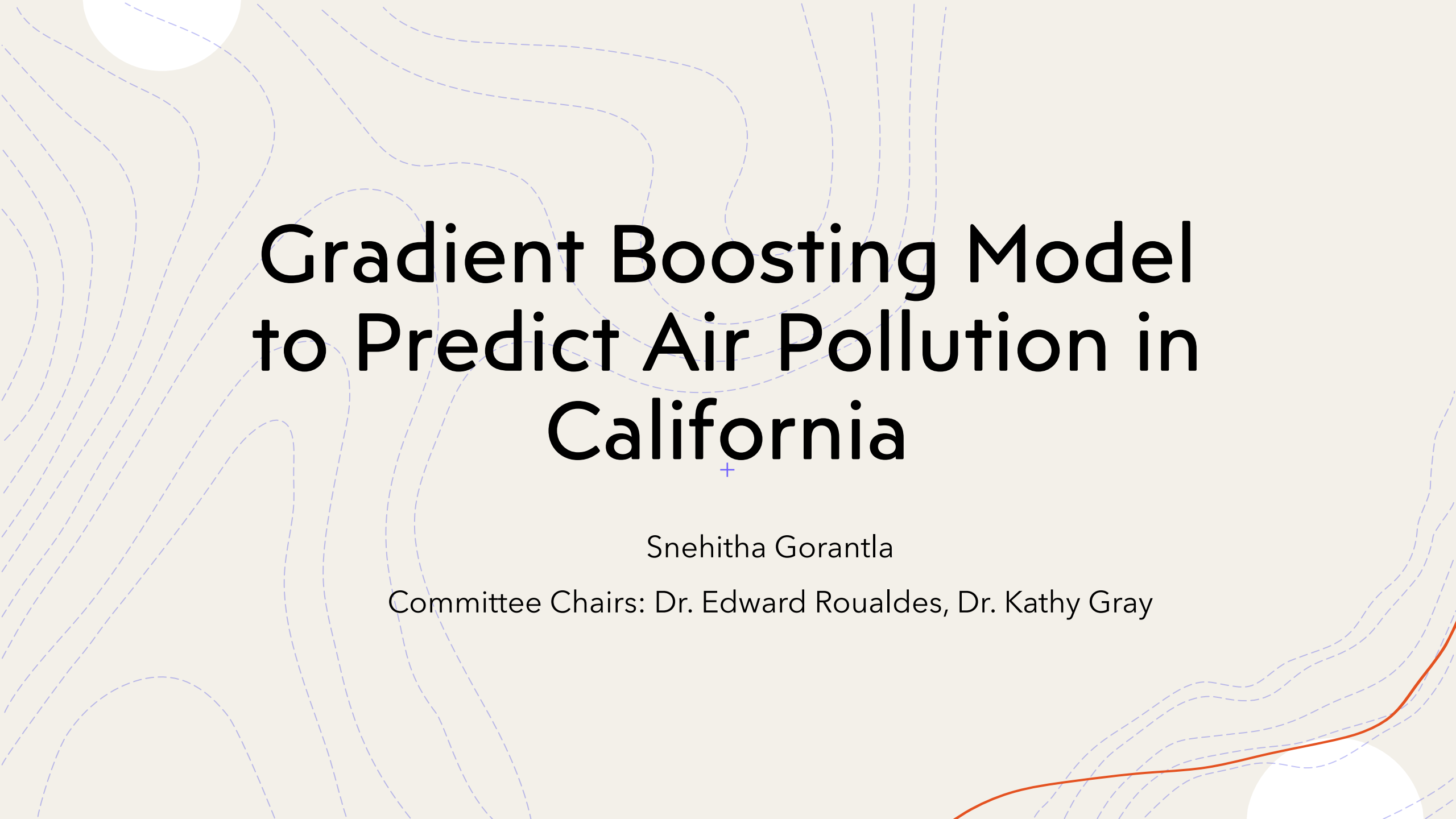


The background features a light beige color with several thin, wavy, dashed lines in a light blue or purple hue. There are also two solid white circles: one in the top-left corner and another in the bottom-right corner. A single solid orange line curves along the bottom right edge of the slide.

Gradient Boosting Model to Predict Air Pollution in California

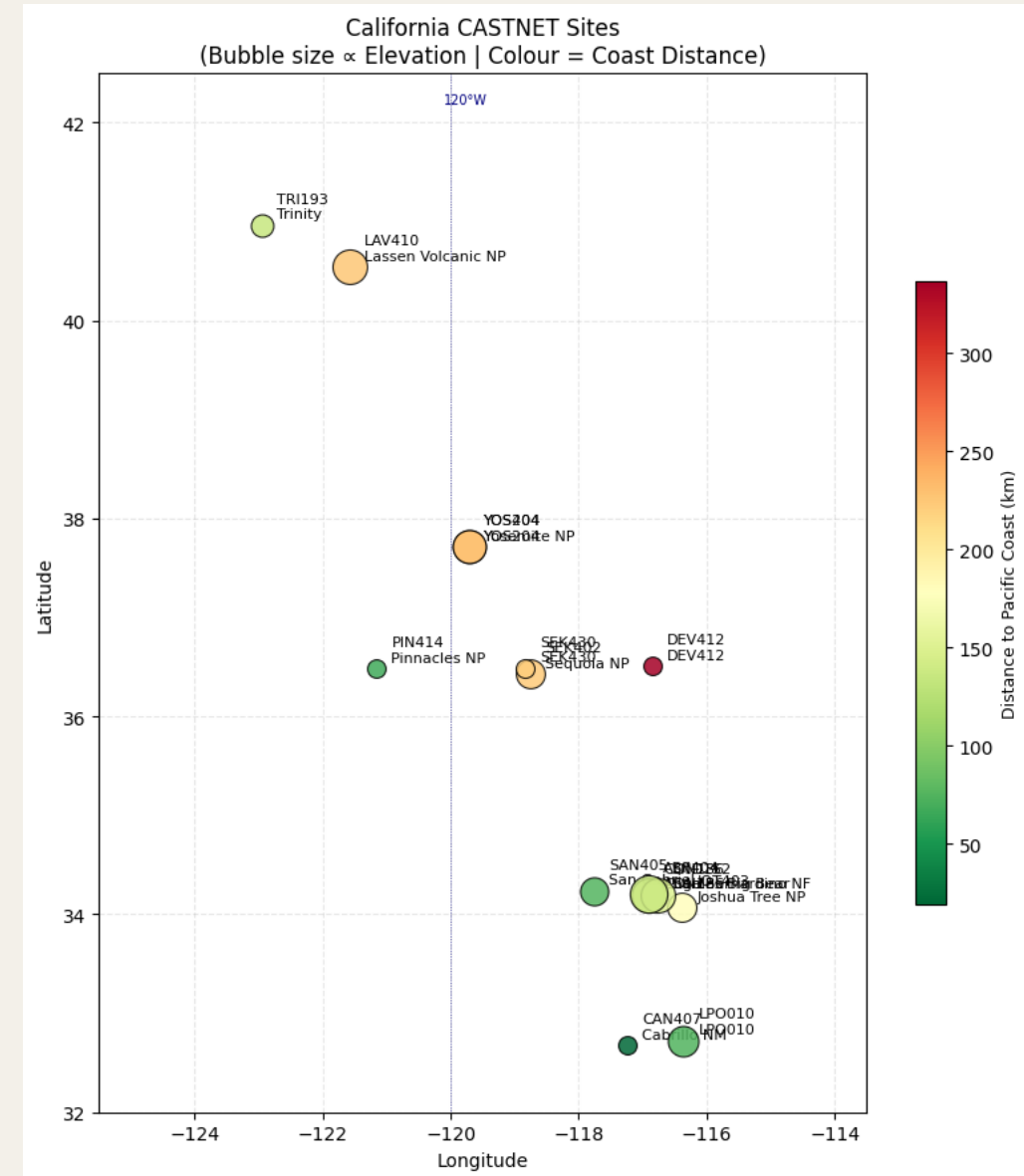
Snehitha Gorantla

Committee Chairs: Dr. Edward Roualdes, Dr. Kathy Gray

Background

The Clean Air Act of 1970 gave the Environmental Protection Agency (EPA) authority to set National Ambient Air Quality Standards (NAAQS) for six criteria pollutants: ozone, particulate matter, carbon monoxide, nitrogen dioxide, sulfur dioxide, and lead.

With the help of NAAQS, ground-level ozone is regulated using the 8-hour rolling maximum.



Data Sources

- + **Primary Source: CASTNET** (Clean Air Status and Trends Network)
 - + **Focus:** Rural and high-elevation background monitoring (National Parks & Wilderness).
 - + **Time Span:** 1990 – 2025 (Continuous hourly records).
 - + **Hourly Gas:** NO, NO₂, NO_y, CO, and SO₂ (Precursor tracking).
 - + **Dry/Wet Chemistry:** Weekly SO₄, NO₃, NH₄ (Deposition & aerosol analysis).
- + **CAL FIRE FRAP** (Fire and Resource Assessment Program)
 - + Data: Historical and active wildfire perimeters.

Literature Review

+ The Wildfire Signature:

- + **Ali et al. (2025):** Established that the 2018 California fires negated years of emission progress, identifying a smoke-enhanced ozone regime.
- + **White et al. (2026):** Demonstrated that hourly $\text{PM}_{2.5}$ estimates from low-cost sensors (PurpleAir) are essential for tracking modern smoke volatility.

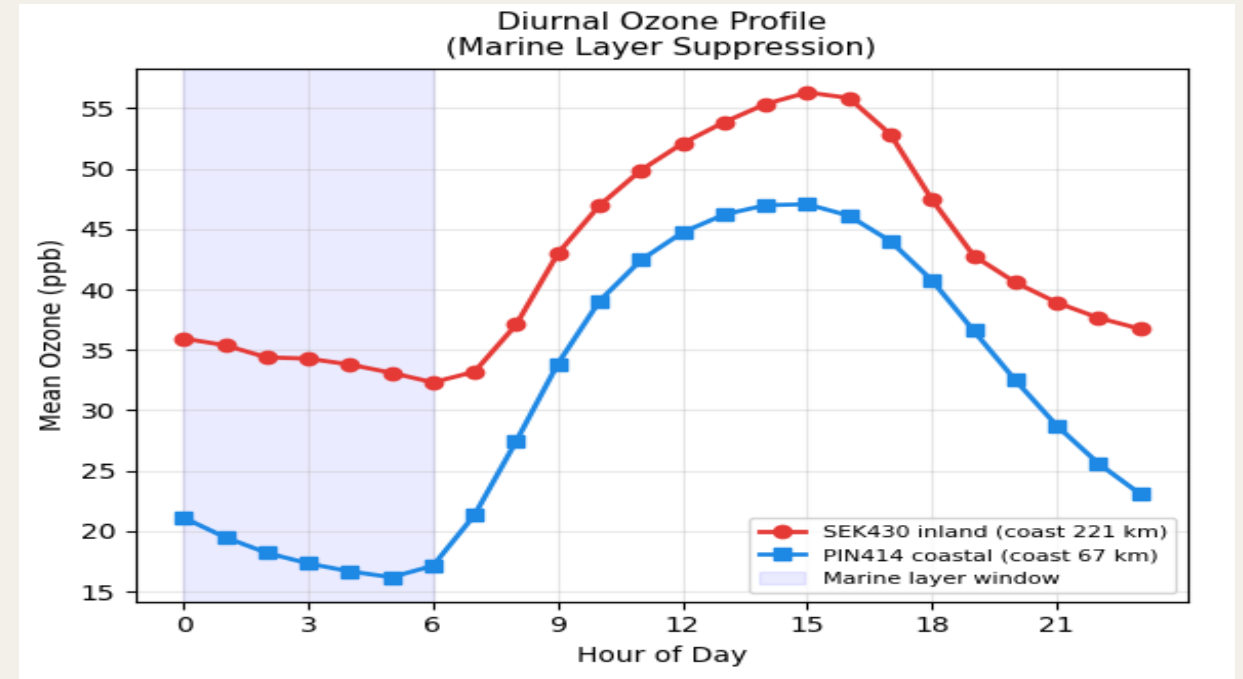
+ The Monitoring Gap:

- + **Orr et al. (2025):** Noted that while we can map smoke perimeters, global models still fail at site-specific predictions in complex California terrain.

+ The Ecological Shift:

- + **Lodge (2025):** Argued for more granular, climate-driven health metrics. This supports our transition from the binary AOT40 index to the continuous W126 metric.

Feature Engineering

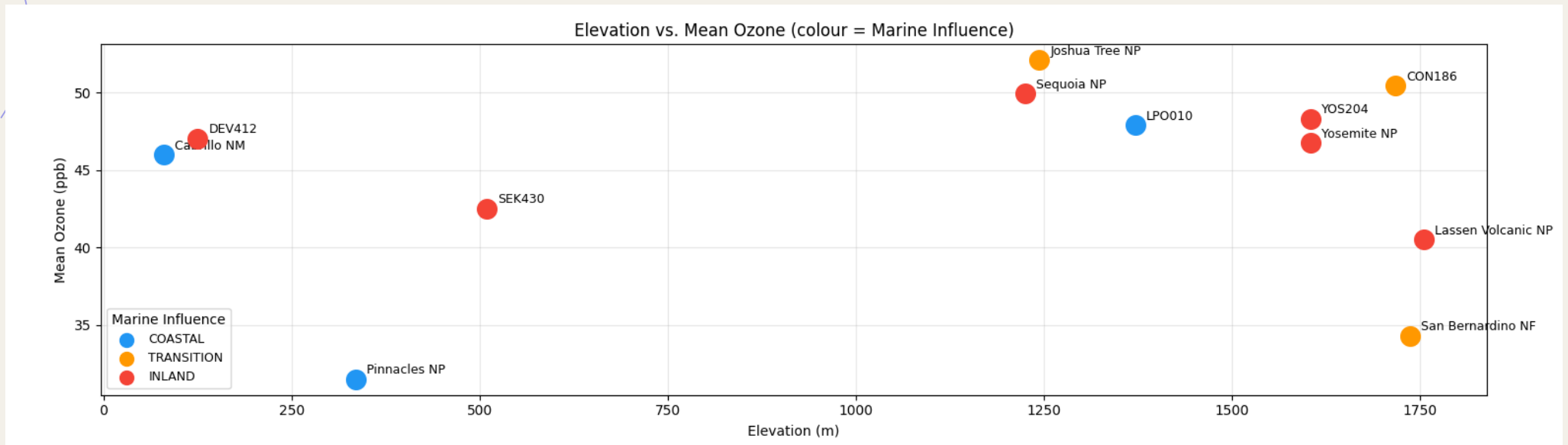


+ Temporal Dynamics & Seasonality:

- + **Within-Day Cycles:** Cyclical encoding for Hour and Day-of-Week to capture the diurnal photochemical cycle.
- + **Seasonal Indicators:** Binary flags for "Peak Summer" and "California Fire Season" to account for shifting baseline emissions.

+ The "Climate Penalty" Proxy:

- + **Post-2017 Indicator:** A binary feature used to differentiate the era of extreme fire and drought from the historical baseline.



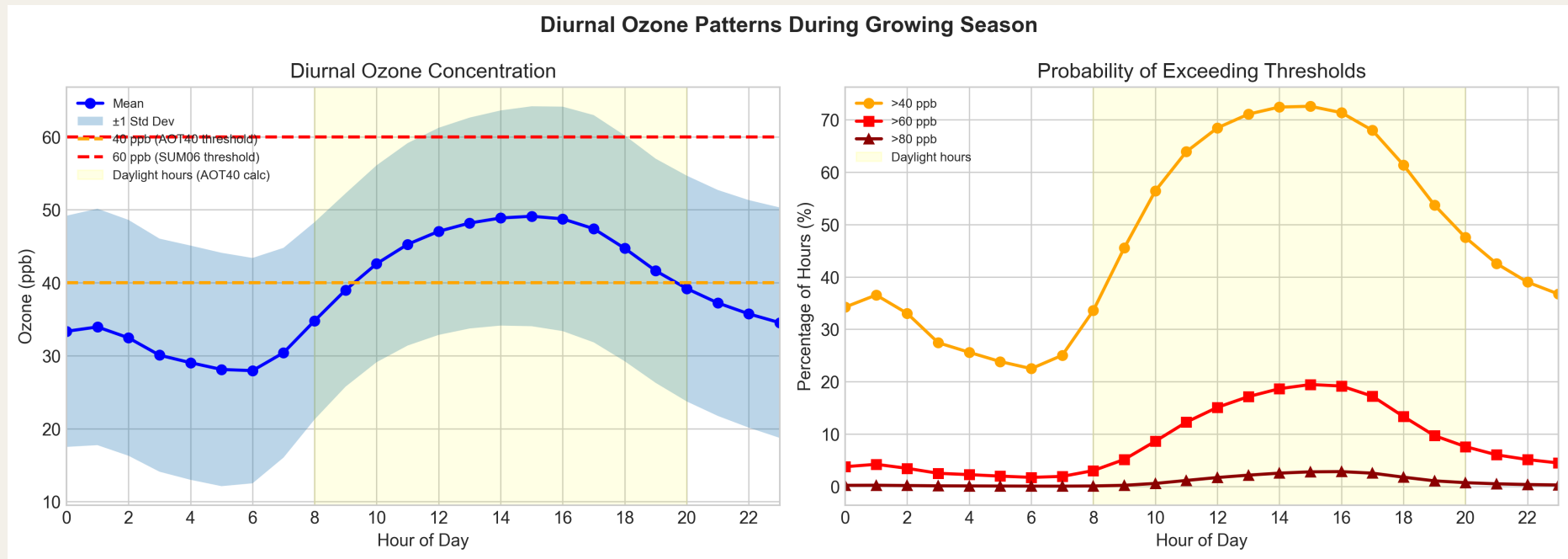
+ Geospatial & Site Characteristics:

- + **Atmospheric Drivers:** Site altitude & slope and Marine Proximity.
- + **Biogenic Factors:** Tree cover and land-use class to account for volatile organic compound (VOC) precursors.

+ Lagged & Rolling Statistics:

- + **Memory Effects:** Rolling window means (3h, 8h, 24h) and autoregressive lags (up to 168 hours) to capture atmospheric persistence.

Feature Engineering



Using the validated ozone predictions, ecological exposure was assessed:

Analysis Window:

- + Growing Season: April-September
- + Daylight Hours: 8 AM – 8 PM

Vegetation Exposure Metrics:

- + AOT40: Accumulated exposure above 40 ppb
- + W126: Sigmoid-weighted index emphasizing higher concentrations

Vegetation Exposure Analysis

Model Training

+ Temporal Partitioning:

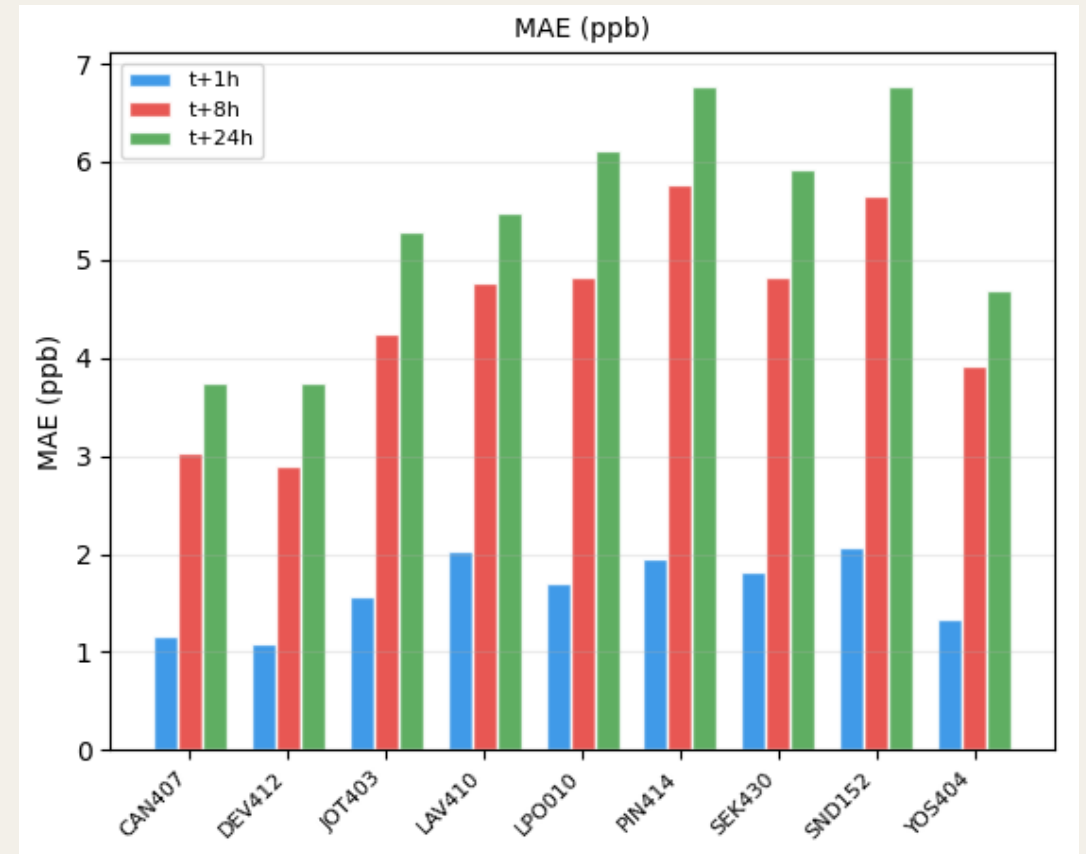
- + **Training (≤ 2018):** 28 years of historical baseline emissions and atmospheric cycles.
- + **Validation (2019-2020):** Specifically chosen for its extreme volatility: the 2020 record wildfire season and the COVID-19 pandemic.
- + **Testing (2021-2025):** Final evaluation on the modern climate and post-pandemic recovery era.

+ Multi-Horizon Strategy: Three distinct LightGBM models trained for t+1h, t+8h (Regulatory), and t+24h (Forecast) horizons.

Results

+ Error Metrics:

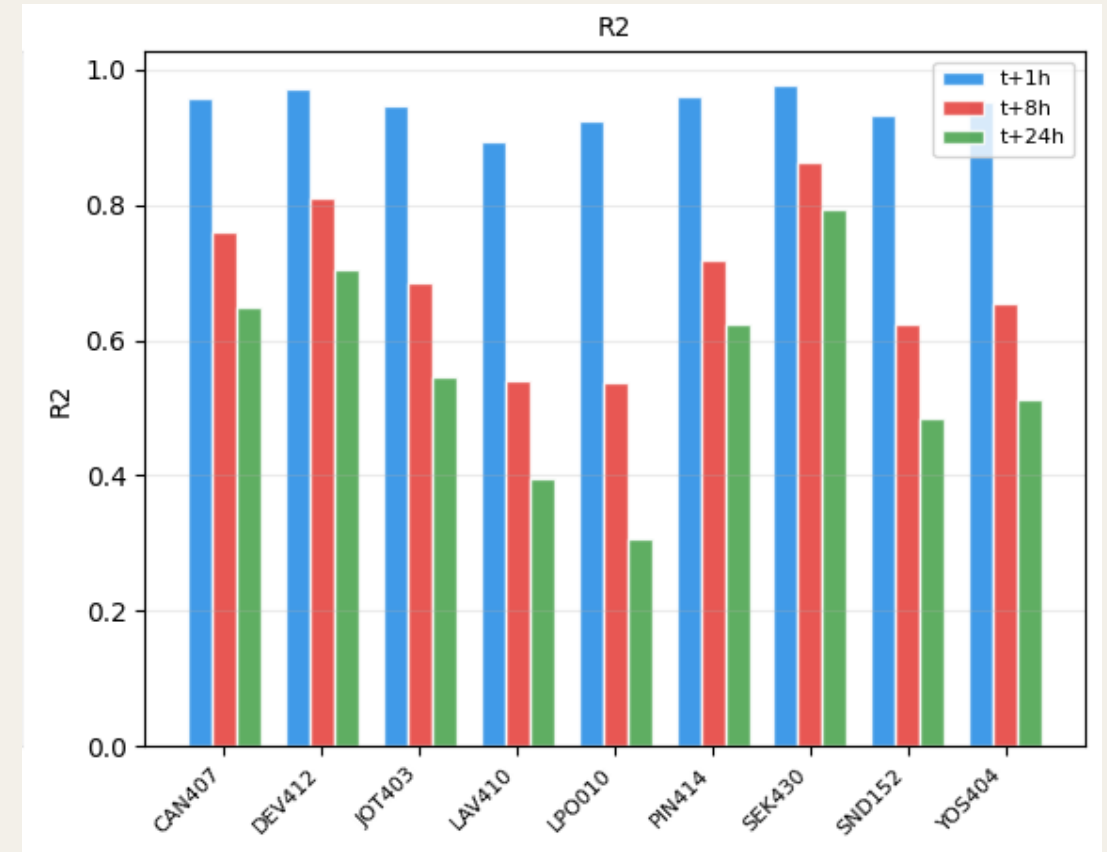
- + **t+1h Excellence:** Near-perfect accuracy with Mean Absolute Error (MAE) hovering between 1-2 ppb.
- + **Horizon Decay:** By t+24h, MAE climbs to 4-7 ppb.



Results

Goodness of Fit:

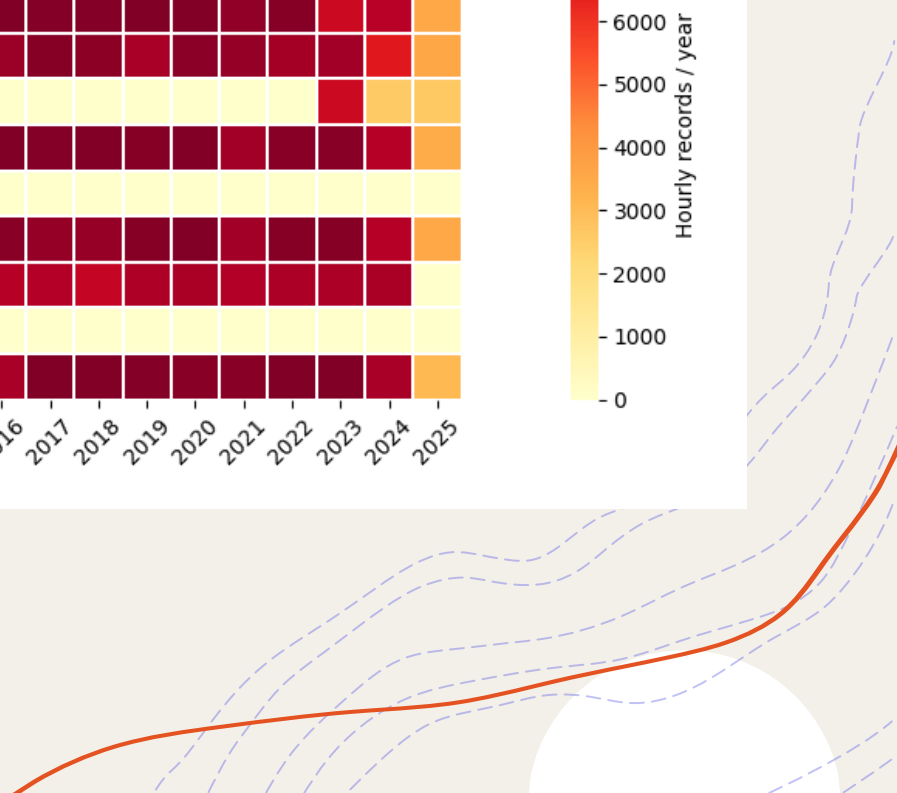
At the t+24h horizon, R^2 varies significantly by geography. Sites like SEK430 remain robust, while LPO010 drops below 0.4, indicating these locations are dominated by stochastic external factors rather than local persistence.



CAN407

CON186

DEV412



Future Scope

- + Using models trained specifically for time series forecasting to better understand the climate patterns.
- + Deploying a real-time interface to track $t+24h$ risk and cumulative W126 stress.
- + Integrating more specialized sensors like PurpleAir to refine the model's understanding of smoke chemistry.

References

- + White, S. R., Sugrue, R. A., Guillotte, L., James, E., Wang, H., Ahmadov, R., ... & Chow, F. K. (2026). Hourly PM_{2.5} Estimates across California from 2018 to 2023. *ACSES&T Air*.
- + Ali, M., Sanyal, S., & Aneja, V. (2025). The Air Pollution Impacts of California's 2018 Wildfires.
- + Orr, A., Adam, C. E., Graham, J., Holden, Z. A., Hu, L., Jaffar, Z., ... & Landguth, E. L. (2025). A state of the science review of wildfire-specific fine particulate matter data sources, methods, and models. *Environmental Health Perspectives*, 133(6), 066001.
- + Lodge, C. (2025). Health effects from climate-driven events: lessons learnt. *Lancet*, 13, 202.

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THANK YOU!